

C++ Programming Tasks & Instructions — CodeAlpha

C++ Programming — Internship Overview

This internship program offers hands-on experience in **C++ programming** and software development. CodeAlpha is a leading software development company committed to innovation, performance, and scalable solutions. The internship empowers students to master core **C++ concepts**, develop efficient algorithms and build high-performance applications. Interns will gain practical skills in memory management, object-oriented programming and debugging with expert mentorship and opportunities to contribute to real-world projects in a collaborative environment.

Internship Perks

- Internship Offer Letter
 - Completion Certificate (QR Verified)
 - Unique ID Certificate
 - Letter of Recommendation (based on performance)
 - Job Opportunities / Placement Support
 - Resume Building Support
-

Instructions for Interns

1. Share your internship status on **LinkedIn**, tagging **@CodeAlpha**.
 2. Complete the **assigned projects** within the mentioned time frame.
 3. Upload your complete source code to **GitHub** in a repository named:
`CodeAlpha_ProjectName`
 4. Post a **video explanation** of your project on LinkedIn with GitHub repo link.
 5. Submit your completed task using the **Submission Form**.
 6. **Complete any 3 or 2 out of the 4 tasks** listed below (from your domain).
-

C++ Programming Task List

(Complete any 2 or 3 of the following tasks)

TASK 1: CGPA Calculator

- Take input for the number of courses taken by the student.
 - For each course, input the grade and the credit hours.
 - Calculate the total credits and total grade points (grade × credit hours).
 - Compute the GPA for the semester and then the overall CGPA.
 - Display individual course grades and the final **CGPA** to the user.
-

✓ TASK 2: Login and Registration System

- Create a registration function that takes **username** and **password** as input.
 - Validate the inputs and check for duplicate usernames if needed.
 - Store user credentials securely in a file (one file per user or a database file).
 - Implement a **login** function that reads credentials and verifies user identity.
 - Provide appropriate success or error messages for **registration** and login.
-

✓ TASK 3: Sudoku Solver

- Represent the **Sudoku grid** as a 2D array.
 - Implement a backtracking algorithm to fill empty cells with valid numbers.
 - Check for the Sudoku rules (row, column, and 3x3 subgrid constraints) before placing a number.
 - Recursively try possible numbers until the puzzle is solved.
 - **(Optional)** Add a GUI for easier user input and display of the solution.
-

✓ TASK 4: Banking System

- Design classes for **Customer**, **Account** and **Transaction** with necessary attributes.
 - Implement functions to create and manage customers and accounts.
 - Include features for **deposits**, **withdrawals** and **fund transfers**.
 - Store transaction history and allow users to view recent transactions.
 - Display account information like balance and **transaction details to customers**.
-



Important Note

Internship Completion Criteria:

To be eligible for the internship certificate, participants must complete a minimum of **two or three tasks**. Submitting only one task will be considered **incomplete** and certificates will **not** be issued in such cases.



Submission Details

A submission form will be shared in your respective **WhatsApp group**. You are required to submit your completed task only through that form. Please follow the instructions mentioned in the form carefully to ensure your submission is accepted.



Contact Information

- Website: www.codealpha.tech
- WhatsApp: [+91 9336576683](https://wa.me/919336576683)
- Email: services@codealpha.tech , services.codealpha@gmail.com